



# SMARTIE

Synergistic Management and Advancement of Artificial  
Intelligence in European Higher Education

## WP2. Alliance for Artificial Intelligence in Higher Education – Setup SMARTIE

*T2.2. Analysis related to adoption AI in educational process*

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*D.2.2.4. Survey among stakeholders*



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## 1. Foreword

The rapid development of artificial intelligence has begun to reshape not only higher education, but also the wider socio-economic environment in which universities operate. Employers, public institutions, non-governmental organizations, professional associations, and other external actors are increasingly confronted with the opportunities and challenges brought by AI-based technologies. These developments affect work processes, decision-making practices, service provision, innovation capacity, and the skills required from present and future employees.

In this context, it becomes essential to understand how stakeholders perceive the integration of artificial intelligence into their own activities and how they assess the role of universities in supporting this transformation. The interaction between higher education institutions and external stakeholders is a key element in ensuring that educational provision remains relevant to labour market developments and to the digital transformation of society. At the same time, the growing importance of AI in professional environments raises new expectations regarding the competences that graduates should acquire during their studies.

## 2. Introduction

Surveying stakeholders is of critical importance for obtaining an evidence-based understanding of the current and future role of AI in professional and organizational practice. External stakeholders represent an essential interface between higher education and the labour market, as they can provide direct insight into the emerging needs of employers, sectoral trends, digital transformation challenges, and expectations regarding graduates' competences. Understanding their perceptions, experiences, and needs is therefore necessary in order to align higher education more effectively with the changing realities of work and innovation.

A structured survey among stakeholders makes it possible to identify the extent to which artificial intelligence is already being integrated into institutional, economic, administrative, or professional activities, as well as the main barriers and support needs associated with this process. It also provides valuable information regarding the forms of collaboration that stakeholders expect from universities in order to facilitate AI adoption, such as training, consultancy, applied research, curriculum adaptation, internships, continuing education, and knowledge transfer.

Moreover, such a survey offers important insight into the relationship between AI-related competences and graduate employability. Stakeholders are in a strong position to assess whether graduates who possess knowledge, abilities, and practical competences in the use of AI can be integrated more effectively into the labour market, and which specific competences are most relevant in this regard. In this way, the survey contributes to a better understanding

of how higher education institutions can modernize their educational offer and strengthen the preparedness of graduates for an increasingly AI-influenced labour market.

This survey was therefore conceived as an essential step in exploring the views and expectations of stakeholders regarding AI integration in their own fields of activity, the support that universities can provide in this process, and the role of AI competences in improving graduate readiness for employment and professional development.

### 3. Methodology of survey

The survey among stakeholders was implemented through an online data collection methodology designed to ensure accessibility, consistency of responses, efficient data management, and straightforward export of results for subsequent statistical analysis. Since the questionnaire had already been developed prior to the technical implementation stage, the methodological focus was placed on the digital construction of the survey instrument, online distribution to respondents, secure collection of responses, and conversion of the collected data into a structured spreadsheet database. For these purposes, Questback was selected as the online survey platform, as it provides a user-friendly interface, robust data security standards, and comprehensive analytics tools that facilitate efficient data collection and insightful reporting. The first methodological stage consisted of transposing the already finalized questionnaire into Questback. In Questback, the steps for creating a questionnaire are organized as follows: Settings, Questionnaire, Languages, Distribution, Notifications, Publish, Results, and Follow-Up.

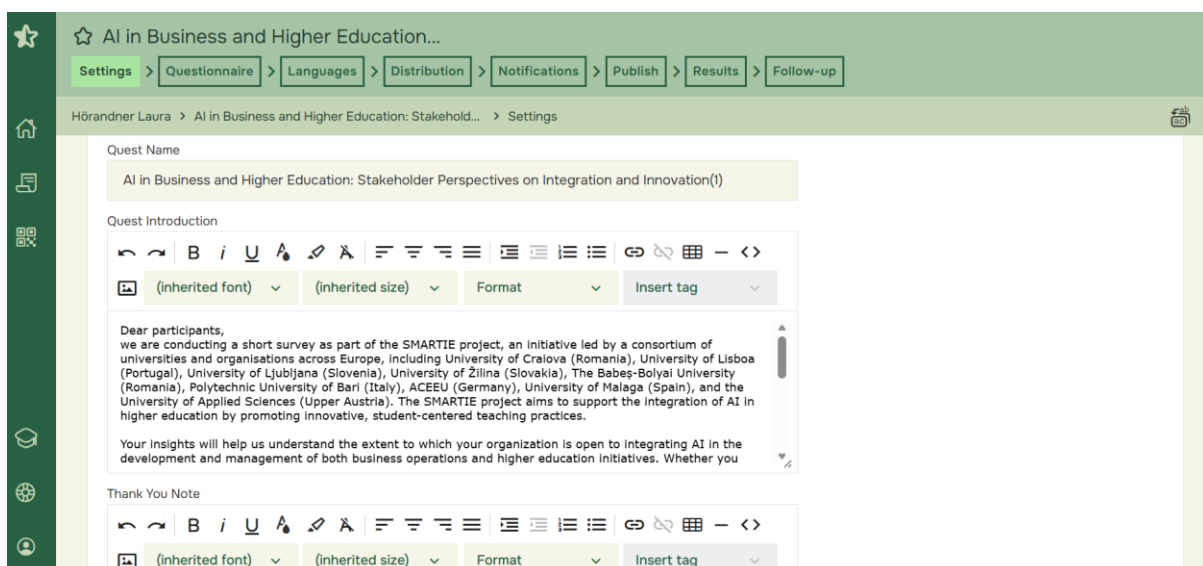


Figure 1: Settings in Questback

After setting up the general settings (such as survey introduction, name, type, hidden identity, and response options), the next step is the questionnaire section, where each question is added individually by defining its type, properties, text, and specific settings.

The questionnaire addressed several core dimensions relevant to stakeholders. First, it explored the extent to which stakeholders were already integrating, or intended to integrate, artificial intelligence into their organizational, administrative, productive, or service-related activities. Second, it examined stakeholder perceptions regarding the role that universities could play in supporting this integration through training, curriculum adaptation, applied research, cooperation initiatives, and knowledge transfer. Third, it investigated the extent to which graduates could be better integrated into the labour market if they possessed relevant AI-related abilities and competences.

Questback allowed the research team to insert questions directly in the editor by selecting the appropriate question type for each question, such single select, matrix, multiple choice, open answer or miscellaneous. This flexibility was particularly useful for a stakeholder survey, where the questionnaire needed to include both factual identification items and attitudinal or evaluative questions. Moreover question properties, question settings, display logic and format can be edited in Questback.

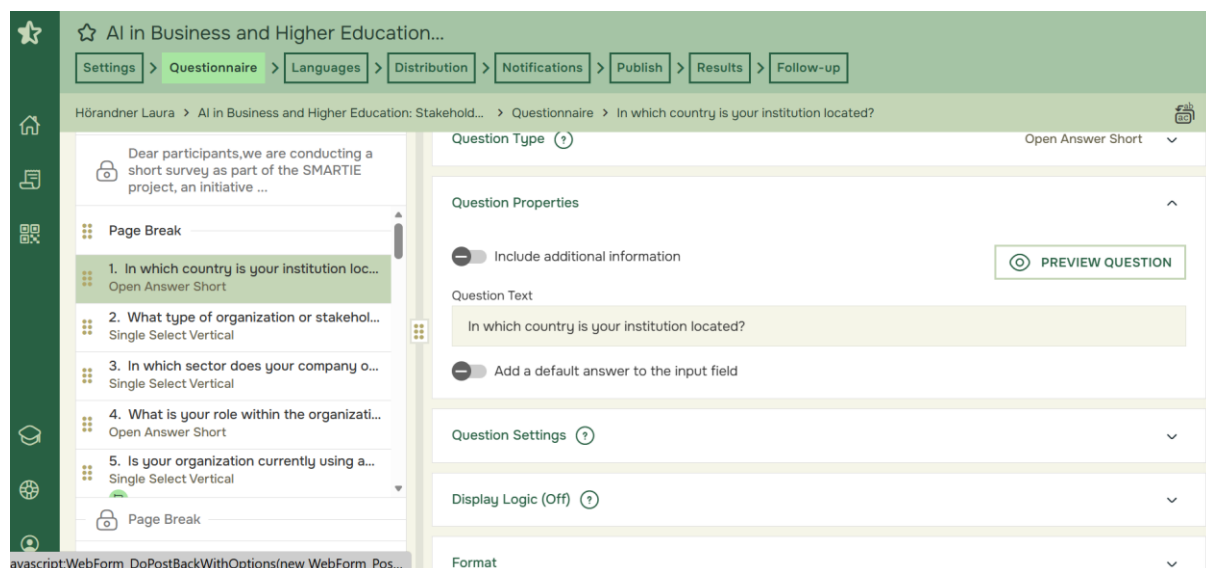
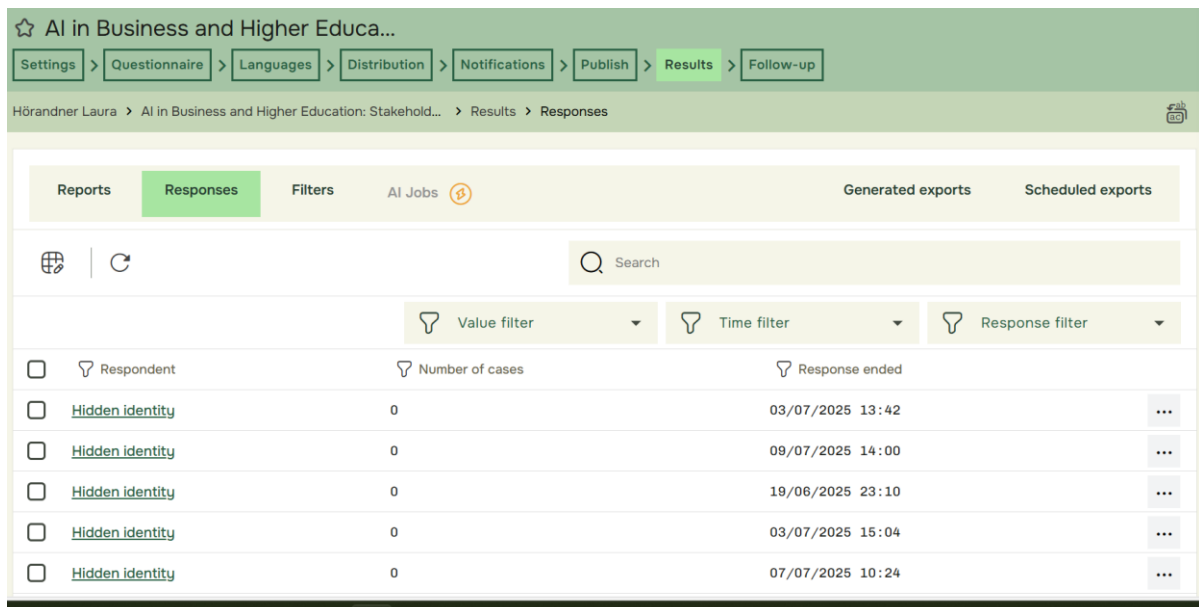


Figure 2: Questionnaire settings in Questback

Once the questions were entered, the next step was the configuration of survey logic and completion rules. In a survey addressed to stakeholders, it is often necessary to maintain coherence between questions and reduce respondent burden. The online platform includes conditional logic that can show or hide specific questions, sections, or pages depending on previous answers. This capability was methodologically important because it improved response quality, limited irrelevant answers, and made the online questionnaire more user-friendly.

After the construction of the form, the methodology proceeded to the publication and distribution stage. Once the survey instrument was finalized in the editor, it had to be published so it could be accessed online by stakeholder respondents. The platform's workflow supports sharing the form after publication and collecting responses through a web link. This method was particularly appropriate for stakeholders distributed across different regions, sectors, and institutional contexts, because it removed the logistical barriers associated with paper questionnaires or face-to-face data collection. The survey link was distributed by email and QR-code and through relevant communication channels to representatives of companies, employers, business support organizations, public institutions, non-governmental organizations, professional bodies, labour market actors, and other external stakeholders connected to higher education and graduate employability. In this way, the same standardized questionnaire was delivered to all respondents, ensuring comparability of the collected data.

Regarding how data were sent and collected from respondents, the methodological flow was straightforward. Each respondent received the survey link and accessed the questionnaire through a browser. After filling in the form, the respondent submitted the completed questionnaire electronically. At that point, the platform recorded the submission in its submissions area, where the research team reviewed all received answers during the analysis stage. The platform stored response data and made them accessible in the results section.



The screenshot shows the 'Results' section of the Questback platform. The breadcrumb trail is: AI in Business and Higher Educa... > Settings > Questionnaire > Languages > Distribution > Notifications > Publish > Results > Follow-up. The current page is 'Results > Responses'. The interface includes a search bar, filters for 'Value filter', 'Time filter', and 'Response filter', and a table of responses.

| Respondent                      | Number of cases | Response ended   |
|---------------------------------|-----------------|------------------|
| <a href="#">Hidden identity</a> | 0               | 03/07/2025 13:42 |
| <a href="#">Hidden identity</a> | 0               | 09/07/2025 14:00 |
| <a href="#">Hidden identity</a> | 0               | 19/06/2025 23:10 |
| <a href="#">Hidden identity</a> | 0               | 03/07/2025 15:04 |
| <a href="#">Hidden identity</a> | 0               | 07/07/2025 10:24 |

Figure 3: Results section in Questback

In terms of content, the survey questions were structured so as to generate evidence in three main directions. First, the survey examined the current state of AI integration in stakeholder activities, including the extent of actual use, planned adoption, perceived benefits, and key barriers. Second, it investigated the forms of support that stakeholders expected from

universities in relation to AI integration, such as tailored training, continuing education opportunities, curriculum modernization, collaboration in research and innovation, student placements, and consultancy. Third, it explored how far graduates possessing AI-related abilities and competences could be integrated more effectively into the labour market, and which specific skills were regarded as most relevant for employability and workplace adaptability.

After data collection, the final methodological component concerned the creation of a spreadsheet database for analysis. The platform provided a native option to download all results in a single CSV file, which was opened directly in Microsoft Excel and saved as an Excel workbook for the creation of an XLS/XLSX database.

| Date                | 1: In which country is your institution located? | 2: What type of organization or stakeholder do you represent? | 3: In which sector does your company operate? | 4: What is your role with |
|---------------------|--------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------|---------------------------|
| 19.06.2025 23:10:38 | Portugal                                         | Research or think tank organization                           | Manufacturing                                 | Project Manager           |
| 30.06.2025 13:43:39 | Germany                                          | Regulatory or accreditation body                              | Education                                     | Digital Solutions Designe |
| 30.06.2025 14:30:39 | Germany                                          | Regulatory or accreditation body                              | Education                                     | Knowledge Coordinator     |
| 30.06.2025 18:25:50 | Portugal                                         | Educational institution (school, university, training center) | Education                                     | Professor                 |
| 03.07.2025 07:52:26 | Austria                                          | Business enterprise (for-profit)                              | Technology                                    | Teamleiter Disposition N  |
| 03.07.2025 10:27:30 | Portugal                                         | Technology or EdTech provider                                 | Technology                                    | Developer                 |
| 03.07.2025 10:42:53 | Portugal                                         | Real estate                                                   | Real estate                                   | Mca - market centee ad    |
| 03.07.2025 10:54:06 | Spain                                            | Educational institution (school, university, training center) | Healthcare                                    | Lecturer                  |
| 03.07.2025 11:06:24 | Spain                                            | Educational institution (school, university, training center) | Healthcare                                    | Dean                      |
| 03.07.2025 11:08:48 | Spain                                            | Educational institution (school, university, training center) | Education                                     | Proffesor                 |
| 03.07.2025 11:15:37 | Portugal                                         | Business enterprise (for-profit)                              | Healthcare                                    | CEO                       |
| 03.07.2025 11:42:25 | Spain                                            | Individual expert or consultant                               | Travel and events                             | Freelance                 |
| 03.07.2025 11:42:45 | España                                           | Technology or EdTech provider                                 | Technology                                    | Arquitecto                |
| 03.07.2025 11:56:28 | Spain                                            | Educational institution (school, university, training center) | Education                                     | Teacher Assistant         |
| 03.07.2025 11:58:08 | Netherlands                                      | Technology or EdTech provider                                 | Finance                                       | Software engineer         |
| 03.07.2025 12:12:21 | Belgium                                          | Educational institution (school, university, training center) | Healthcare                                    | Professor                 |
| 03.07.2025 12:19:46 | Spain                                            | Educational institution (school, university, training center) | Education                                     | Professor and research    |
| 03.07.2025 12:40:38 | Spain                                            | Educational institution (school, university, training center) | Education                                     | Education and research    |
| 03.07.2025 12:42:10 | Portugal                                         | Business enterprise (for-profit)                              | Hospitality                                   | CFO                       |
| 03.07.2025 12:44:34 | Spain                                            | Educational institution (school, university, training center) | Healthcare                                    | Lecturer                  |
| 03.07.2025 13:23:13 | Spain                                            | Educational institution (school, university, training center) | Healthcare                                    | Lecturer                  |
| 03.07.2025 13:42:24 |                                                  | Business enterprise (for-profit)                              | Manufacturing                                 | CEO                       |

Figure 4: Excel file with all results

This Excel file was used to carry out the analysis.

## 4. Conclusion

In conclusion, the survey methodology among stakeholders relied on a fully digital workflow in which an already developed questionnaire was implemented in Questback, configured with appropriate question types and logic, distributed online through a standardized survey link, and completed by respondents via electronic submission. The responses were stored in the platform's submissions environment and were exported for analysis.

This approach ensured standardization of the instrument, ease of access for geographically dispersed stakeholders, efficient collection of data, and compatibility with Excel-based analytical procedures. Overall, the use of Questback provided a practical and methodologically robust framework for administering the stakeholder survey and producing a reliable dataset for subsequent interpretation and reporting. At the same time, the thematic

orientation of the survey made it possible to gather relevant evidence on the extent to which stakeholders integrate AI in their activities, the support they expect from universities in this process, and the importance of AI-related competences for improving graduates' labour market integration. This makes the survey a valuable tool for informing future cooperation between universities and external stakeholders and for strengthening the relevance of higher education in an AI-driven society.